

Structure, Not Loss: Sync-Aware Joint Detection for Semi-Blind Single-Channel Co-Frequency Reception

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Abstract—Single-channel blind source separation of co-frequency communication signals is judged by waveform-fidelity metrics, but a receiver's deliverable is bits. On a controlled benchmark the two decouple structurally: a trained slot separator's outputs, demodulated by an idealised oracle-carrier receiver, track our exact per-constellation separate-detection error floor at two or more sources, while per-slot blind synchronisation is strictly worse by a noise-flat 7–11 points, captured by the residual interferer's spectral line. For such linear-residual separate-then-detect receivers, waveform-level fidelity alone cannot eliminate this floor at reachable separator quality. The remedy is structural: a joint maximum a posteriori detector whose union bound tracks simulation through the sum-constellation minimum distance; an identifiability analysis showing only the effective per-source coefficients are recoverable, up to finite rotational symmetry, through the joint discrete structure; and a Fisher analysis showing blind frequency acquisition is information-poor exactly at co-frequency. A learning-free blind carrier synchroniser matches the oracle for phase-shift-keyed bursts, and a training-free joint synchroniser-detector on the raw two-source mixture with no oracle carriers, phases or symbols ties the strongest waveform route at the paper's separator quality (0.531 vs. 0.534 pooled error rate), winning beyond 5dB; a 26× larger published separator lifts that route to 0.444, yet at matched oracle frequencies the joint architecture still wins (0.411 vs. 0.444). The semi-blind receiver's envelope is priced honestly: blind modulation-pair selection and three-source acquisition fail, and the separator's count head is unreliable. Separation serves slot generation; detection belongs on the mixture, jointly with synchronisation.

Index Terms—Blind source separation, co-frequency communication, carrier synchronisation, joint detection, symbol detection, identifiability, Cramér–Rao bound.

I. Introduction

The problem. Single-channel blind source separation (SC-BSS) of co-frequency communication signals—several modulated emitters overlapping in time and frequency at one receiving antenna—has progressed through a sequence of learned separators trained and ranked by waveform-fidelity metrics [1], [2], [3], [4], [5], [6], [7]. But the deliverable of a communication receiver is bits, not waveforms, and the two need not move together: what remains in a separated stream is residual interference, whose structure—not its energy—determines the demodulated symbol error rate (SER). This paper demonstrates the mismatch, explains it, and fixes it on one controlled variable-count benchmark (Section III-A), starting with our own first experiment (E3, Section V-C): a trained slot separator's outputs, demodulated by an idealised compensated receiver with oracle carriers and phases, sit at SER 0.553/0.586 for $K=2/3$ — just above the exact separate-detection floor we derive in Section III-B (1/2 for

QPSK pairs at SIR = 0dB) and far above the $K=1$ level of 0.145 — and replacing the oracle receiver by a realistic blind synchroniser makes things strictly worse, by an SNR-flat 8–11 points at $K \geq 2$. The explanation is structural: independent per-stream synchronisation is captured by the residual interferer's spectral line (2–4Hz of bias at every SNR), and separate-then-detect discards the linear superposition structure that joint detection exploits. The remedy is derived, not searched.

This paper. Three theoretical findings organise the design, each validated by simulation before the end-to-end system is evaluated. (i) At SIR = 0dB, joint maximum a posteriori (MAP) detection of a symbol pair from the mixed symbol stream is governed by the minimum distance $d_{\min}(\varphi)$ (5) of the sum constellation $C_1 + e^{j\varphi}C_2$, which is strictly positive for almost every relative phase φ ; separate-then-detect, by contrast, has an exact, computable, SNR-independent error floor (Section III-B). (ii) Under continuous-symbol priors the per-source carrier phases of a co-frequency superposition are not identifiable from the mixture; discrete alphabets collapse the continuous gauge freedom to the finite group $\mathbb{Z}_{M_1} \times \mathbb{Z}_{M_2}$ (extended by S_2 for equal alphabets), so the effective complex coefficients are generically identifiable — away from constellation-collision rotations — only through the joint discrete structure (Section III-C). (iii) Blind per-source frequency estimation from the mixture is information-poor when the two carrier offsets differ by less than the inverse burst duration: the exact Fisher inflation of the single-tone Cramér–Rao bound (CRB) has median $1.70\times$ and 90th percentile $7.6\times$ under the benchmark's offset law — rising to median $4.6\times$ and 90th percentile $287\times$ with unknown gains — while the full-waveform Fisher analysis shows no two-source coupling: the disambiguating information is carried by the modulation-bearing waveform, but exploiting it requires symbol-aware or decision-directed processing (Section III-D). The experiments then test the module order this suggests — separate, per-slot sync, joint re-detection — and refine it: per-slot sync is captured by the interferer at $K \geq 2$, and the working receiver synchronises and detects jointly (Section V).

Contributions.

- 1) Theory with simulation validation. A joint-MAP union bound and an exact discrete-interferer separate-detection floor for the symbol-rate mixture model (Theorems 1 and 2), the $d_{\min}(\varphi)$ landscape of PSK and QAM sum constellations, phase-identifiability propositions, an exact two-tone frequency CRB inflation analysis, and its communication-waveform Fisher counterpart (known symbols, unknown channels) that together locate the frequency-disambiguating information in the modulation-bearing waveform — reachable only through symbol-aware or decision-directed processing — each accompanied by a Monte-Carlo or measurement validation (Section V-B).
- 2) BlindCarrierSync. A classical, learning-free burst carrier synchroniser for the benchmark's short bursts

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- (256 symbols, residual carrier offset $[-5, 10]$ Hz) that matches an oracle-carrier receiver within 0.6 SER points at $\text{SNR} \geq -5$ dB for BPSK/QPSK and within 2–4 points for 8PSK (Section V-A). Its design was itself measurement-driven: the textbook decision-directed phase refinement is shown to be fold-biased under fading and is removed (Section IV-B).
- 3) JointPairDetector. Joint ML re-detection of slot pairs over the $\leq 16 \times 16$ hypothesis grid with the 2×2 complex coupling matrix estimated blind by a restart-robustified EM procedure (Section IV-D).
 - 4) End-to-end evaluation (E3–E6). The full pipeline on the benchmark of Section III-A: E3 demonstrates the waveform-to-bit gap directly — the separator’s outputs under an oracle receiver track the exact per-constellation separate-detection floor at $K \geq 2$, and blind per-slot sync loses an SNR-flat 7–11 SER points, tying a no-separation baseline; E4 shows the joint ECM synchroniser-detector on the raw $K=2$ mixture — per-source ISI taps estimated inside the same loop — ties the strongest waveform route at the paper’s separator quality (separate, oracle-synced, PSP/Viterbi: pooled SER 0.531 vs. 0.534) with no oracle carriers, phases or symbols during inference, winning beyond 5 dB — robustly to separator strength: the same route built on a $26\times$ larger published separator (CNSE) reaches 0.444 only through its oracle synchroniser (blind: 0.536) and trails the joint loop at matched oracle frequencies (0.411 vs. 0.444); E5 prices the envelope (blind modulation-pair selection and the tested $K=3$ acquisition fail, for measured reasons) and E6 sweeps the carrier-separation, timing and amplitude boundaries (Sections V-C–V-F).
 - 5) RS-SlotSepNet. The receiver-structured separator combining these modules with our slot-separator baseline (Section IV); the E4 outcome assigns it the slot-generation/localisation role rather than the detection front-end (its learned count head is unreliable at $K=2$; Section V-C).

The remainder of the paper is organised as follows. Section II reviews related work. Section III develops the theory with its validations. Section IV describes the architecture. Section V reports experiments E1–E6 in full. Section VI discusses, Section VII states limitations, and Section VIII concludes.

II. Related Work

A. SC-BSS for communication signals

Blind separation of co-frequency communication signals from a single sensor violates the overdetermined requirement of classical ICA; classical workarounds exploit structure, and the setting has driven a line of learned separators: convolutional [1], recurrent [3], attention-augmented [4], complex-domain [2], spatial-temporal [5] and state-space [6] designs, recently extended to an unknown source count by coupling separation with counting [7] — the family our benchmark (Section III-A) and baseline separator (Section IV-C) belong to. Crucially, several of these works already reach past the waveform: Hou and Gao [1] couple their convolutional separator to an LC-PSP demodulator and report symbol error rates; Chen et al. [3] recover information bits directly with a bidirectional recurrent network; and hybrid knowledge–data designs train the separator with a demodulation-aware loss [9], while long-sequence state-space separators target the same waveform-recovery front-end [10]. Receiver-aware SC-BSS is therefore not new — what is missing is a

quantified account of when and why the separate-then-detect architecture breaks. We measure that cost exactly (Section V-C): on a controlled variable-count benchmark, a separator’s outputs under an idealised receiver track the exact per-constellation separate-detection floor we derive, and the bottleneck is the missing receiver structure rather than the separator’s waveform fidelity within the quality range we can reach.

B. Task-oriented communication and its limits

Task-oriented communication argues that processing should be optimised for the consumer of the message rather than for reconstruction fidelity [12], [13]; learned end-to-end transceivers [14] follow it. Applied to separation, the same principle suggests training the separator on a differentiable demodulation loss — the demodulation-aware objective of the hybrid knowledge–data separator [9] is a recent instance. We take the complementary route — keep the waveform-trained separator, add classical receiver structure — and E3’s oracle arm shows what the loss-level route faces: even ideally synchronised, separate detection of the separated streams tracks Theorem 2’s per-constellation floor at $K \geq 2$, so the bits are bounded by receiver structure, not by separator fidelity alone.

C. Multiuser detection and co-channel receivers

That joint multiuser detection outperforms separate-then-detect pipelines is classical [15], and practical co-channel systems exploit exactly that structure: paired-carrier multiple access (PCMA) lets two transmitters share one carrier and separates them at the receiver with explicit synchronisation, channel estimation and self-interference cancellation [16]. Closest to our mixture-route arms, Yu et al. [8] derive the sufficient statistics of the joint maximum-likelihood blind detector for the single-channel co-frequency symbol-stream model and bound its blind-demodulation performance; our V1/V3 detectors (Section V-D) instantiate the same joint-ML principle, but with the carrier offsets and the 2×2 coupling matrix estimated blind inside the loop rather than assumed — which is what allows the end-to-end, no-oracle evaluation they do not pursue. Neural approximations of multiuser detectors [17], [18] assume symbol-synchronous, equalised inputs — the synchronisation problem is solved before the network sees the signal. We keep the learned separator but reinstate the classical receiver stages around it, and our theory quantifies both detection regimes analytically (Section III-B).

D. Carrier synchronisation and frequency-estimation theory

Non-data-aided carrier recovery for PSK bursts descends from the M -th-power phase estimator of Viterbi and Viterbi [19], and frequency estimation of tones in noise is bounded by the Rife–Boorstyn CRB [20]. We reuse both classical tools, with two benchmark-specific findings: the M -th-power spectral-line estimator must operate on the matched-filtered signal to survive transition-slew spurs (Section IV-B), and the two-tone CRB inflates sharply exactly in the offset range of co-frequency mixtures (Section III-D) — while the full-waveform Fisher analysis (Remark 1) shows the modulation structure itself carries the coupling-free information.

Positioning. Against this literature the contribution is not joint detection as such — that principle is classical — but the quantified account of why the separated route cannot reach it, plus a receiver closing the gap without

oracles: (i) an exact separate-detection floor under a discrete interferer (Theorem 2); (ii) finite-symmetry identifiability of the effective per-source coefficients (Proposition 2); (iii) the two-tone Fisher coupling of carrier-only acquisition with its full-waveform contrast (Proposition 3, Remark 1); (iv) an empirical architecture diagnosis of separate-then-per-slot-sync (E3); and (v) a training-free joint synchronisation-detection loop operationalising the analysis (E4).

III. System Model and Theory

A. Mixture model and symbol-rate reduction

We observe a single complex baseband mixture of K co-frequency sources,

$$y(t) = \sum_{k=1}^K w_k s_k(t) + n(t), \quad t = 0, \dots, T-1, \quad (1)$$

in our benchmark: each source is a burst of $N=256$ symbols from $m_k \in \{\text{BPSK}, \text{QPSK}, \text{8PSK}, \text{16QAM}\}$ (unit-power constellations), root-raised-cosine shaped (roll-off 0.35, 16 samples/symbol), up-converted to a carrier $f_k = f_0 + u_k + \epsilon_k$ with $f_0 = 2$ kHz, $u_k \sim \mathcal{U}(0, 5)$ Hz and in-generator jitter $\epsilon_k \sim \mathcal{U}(-5, 5)$ Hz, passed through a 3-tap multipath fading channel, and mixed with weights $w_k \sim \mathcal{U}(0.4, 0.6)$ (so $\text{SIR} \approx 0$ dB by construction) and AWGN; $T = 4096$ samples at $f_s = 16$ kHz, i.e. a burst duration $T_{\text{burst}} = 0.256$ s; the number of sources varies per mixture, $K \in \{1, 2, 3\}$ in training and on the test grid. Timing is fixed at the zero-delay symbol grid by construction, so timing recovery is out of scope; carrier frequency and phase recovery is not.

For the analysis of detection we use the symbol-rate model that the receiver of Section IV produces after per-slot synchronisation. Consider one mixed symbol stream in which two unit-power symbol sequences $c_{1,n} \in \mathcal{C}_1$, $c_{2,n} \in \mathcal{C}_2$ are superimposed with complex gains:

$$r_n = a_1 c_{1,n} + a_2 c_{2,n} + w_n, \quad w_n \sim \mathcal{CN}(0, \sigma^2). \quad (2)$$

We write $a_1 = 1$ without loss of generality (scale absorbs into σ^2 and $|a_2|$; the reference phase absorbs into $\varphi = \angle(a_2/a_1)$), $\text{SIR} = 1/|a_2|^2$, and $\text{SNR} := 1/\sigma^2$ per unit-power symbol.

B. Joint MAP detection and the separate-then-detect floor

Under known (a_2, σ^2) and uniform symbol priors, the joint MAP/ML detector of the pair from (2) is

$$(\hat{c}_1, \hat{c}_2) = \arg \min_{(c_1, c_2) \in \mathcal{C}_1 \times \mathcal{C}_2} |r - c_1 - a_2 c_2|^2, \quad (3)$$

a search over at most 16×16 hypotheses. Its error probability is governed by the sum constellation $\mathcal{C}_1 + a_2 \mathcal{C}_2$.

Theorem 1 (Joint-detector union bound). For model (2) at fixed φ , the per-symbol error rate of stream 1 under joint ML detection satisfies

$$P_{e,1}(\varphi) \leq \frac{1}{M_1 M_2} \sum_{(c_1, c_2)} \sum_{\substack{(c'_1, c'_2) \\ c'_1 \neq c_1}} Q\left(\frac{|\Delta c_1 + a_2 e^{j\varphi} \Delta c_2|}{\sqrt{2\sigma^2}}\right), \quad (4)$$

with $\Delta c_k = c_k - c'_k$, $M_k = |\mathcal{C}_k|$, and Q the Gaussian Q-function; the bound for stream 2 is analogous (mask $c'_2 \neq c_2$), and the pair-averaged bound is their mean. The bound holds pointwise in φ , so integrating it over any phase ensemble upper-bounds the corresponding phase-averaged error; a finite phase grid only approximates that

integral, with a construction-dependent bias (downward near the zero contacts of Section III-B).

The proof is deferred to Appendix A.

The bound is controlled by the minimum distance of the sum constellation,

$$d_{\min}(\varphi) = \min_{(\Delta c_1, \Delta c_2) \neq (0,0)} |\Delta c_1 + e^{j\varphi} \Delta c_2|, \quad (|a_2| = 1). \quad (5)$$

Because the difference sets of PSK/QAM alphabets are lattices with rational ratios, exact zero contacts $d_{\min}(\varphi) = 0$ exist only at commensurate rotations — for QPSK exactly $\varphi \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ (only equal-magnitude difference pairs can cancel; the diagonal-diagonal cancellations land on the same four angles) — a measure-zero set (Fig. 4). Quantitatively, however, the typical minimum distance is well below the single-user distance d_{su} : the median of $d_{\min}(\varphi)$ over $\varphi \sim \mathcal{U}[0, 2\pi]$ is $0.39 d_{\text{su}}$ for QPSK and $0.16 d_{\text{su}}$ for 16QAM, and for 16QAM 30% of rotations have $d_{\min} < 0.1 d_{\text{su}}$ (Fig. 4). Joint detection at $\text{SIR} = 0$ dB therefore does not reach the single-user bound — it pays a bounded effective-SNR penalty (≈ 8 dB for QPSK at the median rotation) — while separate detection pays an unbounded one, as the next result makes exact.

Theorem 2 (Exact separate-detection SER and its high-SNR floor). Let stream 1 be detected separately, $\hat{c}_1 = \arg \min_{c \in \mathcal{C}_1} |r - c|^2$, treating $a_2 c_2$ as noise. For constellations with axis-aligned decision boundaries (QPSK, square QAM), let $\{\ell_i\}_{i=1}^L$ be the per-axis levels of the unit-power alphabet with boundaries $b_i = (\ell_i + \ell_{i+1})/2$, and let $u = \text{Re}(a_2 e^{j\varphi} c_2)$ be the interferer's per-axis offset. Conditioned on (c_2, φ) , the per-axis error probability is exactly

$$e(u) = \frac{1}{L} \sum_{i=1}^L \left[Q\left(\frac{\ell_i + u - b_{i-1}}{\sigma_r}\right) \mathbf{1}_{i>1} + Q\left(\frac{b_i - \ell_i - u}{\sigma_r}\right) \mathbf{1}_{i<L} \right], \quad (6)$$

with $\sigma_r^2 = \sigma^2/2$ the per-axis noise variance, and the separate-detection SER is $P_{e,\text{sep}} = \mathbb{E}_{c_2, \varphi} [1 - (1 - e(u_I))(1 - e(u_Q))]$, computable by averaging (6) over the M_2 interferer symbols and any phase ensemble. Its high-SNR limit depends on the ensemble: at $\text{SIR} = 0$ dB, as $\sigma^2 \rightarrow 0$,

$$P_{e,\text{sep}} \rightarrow \begin{cases} \frac{1}{2}, & \text{QPSK pairs, } \varphi \sim \mathcal{U}[0, 2\pi], \\ 0.838, & \text{16QAM pairs, } \varphi \sim \mathcal{U}[0, 2\pi], \end{cases} \quad (7)$$

the QPSK value in closed form (Appendix A), the 16QAM value by numerical integration of (6). Two finite effects pull any practical evaluation below the limit: the floor at the zero-contact rotations is lower (7/16 for QPSK), so a phase grid containing the contacts biases the average down, and at finite σ^2 the Q -transition smears a neighbourhood of each contact — at 20 dB the 32-point phase-grid evaluation of (6) gives 0.488 (0.479 with 12 points), below the continuous-ensemble limit (7). Modelling the interferer as Gaussian of the same power (the classic approximation) predicts a QPSK floor of only $1 - (1 - Q(1))^2 \approx 0.292$; the discrete constant-modulus interferer is substantially more damaging, and (6) matches Monte-Carlo simulation to within 0.0036 at every grid point we tested (Section V-B).

Validation. Monte-Carlo simulation ($10^6 / 5 \times 10^5$ symbols per point for QPSK/16QAM pairs, 12 fixed phases offset from the zero contacts; full curves in the public code) compares (4) and Theorem 2 against simulation (the bound is verified pointwise wherever the MC has ≥ 10 error events). Three observations organise the result. First, at typical rotations the joint detector tracks the

union bound within a factor of two (QPSK, SIR = 0 dB, 15 dB SNR: median- φ MC 0.0143 vs. bound 0.0156), while separate detection is pinned at its floor. Second, the phase mean of the joint SER is dominated at high SNR by small neighbourhoods of the zero contacts (QPSK, 20 dB: mean 0.032 vs. median 10^{-4}) — the measure-zero caveat of the $d_{\min}$ statement is a finite-SNR effect of measurable size. Third, for 16QAM pairs the sum constellation is too dense for the burst SNR range (median $d_{\min} = 0.16 d_{\text{su}}$): joint detection still beats separate (0.36 vs. 0.82 at 20 dB) but both are unusable, so end-to-end gains at $K=2$ should be expected to concentrate on PSK pairs. Two boundary behaviours matter for operating the system. The decision rule must match the metric: joint ML optimises the pair, so its per-symbol decisions at SNR $\lesssim 5$ dB are slightly worse than separate detection's (QPSK pairs, SIR = 0 dB, 0 dB SNR: joint 0.479 vs. separate 0.447); the per-source-optimal Bayesian marginal MAP rule removes the reversal (0.446 at 0 dB) and coincides with joint ML wherever the posterior concentrates (SNR ≥ 5 dB, differences < 0.003) — E4 scores both (Section V-D). And at SIR $\gtrsim 15$ dB the weaker stream becomes undecodable from the scalar mixture — symmetric interference is genuinely the sweet spot of joint detection (the end-to-end amplitude-ratio sweep is Fig. 5(d)).

C. Phase identifiability under co-frequency superposition Proposition 1 (Gauge freedom). Let $y = e^{j\phi_1}h_1x_1 + e^{j\phi_2}h_2x_2$ with continuous-amplitude sources. For any (θ_1, θ_2) the reparametrisation $x_k \mapsto e^{j\theta_k}x_k$, $\phi_k \mapsto \phi_k - \theta_k$ leaves y invariant; hence the per-source phases (ϕ_1, ϕ_2) are not identifiable from the mixture under continuous-symbol priors.

The proof (one substitution) is in Appendix A.

Proposition 2 (Discrete alphabets: finite-symmetry identifiability of the effective coefficients). Let $r_n = a_1c_{1,n} + a_2c_{2,n} + w_n$ at symbol rate with $c_{k,n}$ i.i.d. uniform over a known constellation $\mathcal{C}_k$ of rotational symmetry order M_k (M_k -PSK or square QAM), and declare $(a_1, a_2) \sim (\tilde{a}_1, \tilde{a}_2)$ iff $\tilde{a}_k = a_k e^{j2\pi q_k/M_k}$ for integers q_k — or, when $\mathcal{C}_1 = \mathcal{C}_2$, the source swap thereof — so the ambiguity set is the finite group $G = \mathbb{Z}_{M_1} \times \mathbb{Z}_{M_2}$ (resp. its S_2 extension). Then (i) every element of G is an exact symmetry of the mixture law; (ii) conversely, if the sum constellation is collision-free (the map $(c_1, c_2) \mapsto a_1c_1 + a_2c_2$ is injective) and (a_1, a_2) avoids a finite exceptional algebraic set (collision rotations, moment-system degeneracies, modulus ties), then any $(\tilde{a}_1, \tilde{a}_2)$ inducing the same mixture law satisfies $(\tilde{a}_1, \tilde{a}_2) \sim (a_1, a_2)$: the effective per-source complex coefficient — the product of channel response, carrier phase and mixing gain — is identifiable from the population mixture law up to G , and only through the joint discrete structure, since per-slot the rotated constellation is itself a valid hypothesis. Identifiability is not global: at constellation-collision rotations (the zero contacts $d_{\min}(\varphi) = 0$ of Theorem 1, e.g. $\varphi \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ for QPSK+QPSK at equal powers) distinct symbol pairs produce identical mixture points and the conclusion fails. The statement is therefore one of generic, finite-symmetry identifiability of the effective coefficients, not unconditional source or phase recovery.

The proof (moment inversion; Appendix A) is certified numerically for all benchmark constellation pairs, explicitly including the benchmark's SIR ≈ 0 dB unit-amplitude-ratio point (theory_identifiability.py).

Proposition 2 is the theorem-level reason that per-slot frequency information alone cannot suffice in the co-

frequency regime: away from the measure-zero collision set, the effective coefficients are recoverable only jointly with detection — a statement our V0 arm confirms experimentally (Section V-D), and the collision phases themselves are where Theorem 2's floor and the union bound's worst cases live. Two qualifications keep the claim precise. First, $K \geq 2$ alone is not sufficient: as Proposition 3 shows, the estimation coupling vanishes once the offset separation exceeds a few times $1/T_{\text{burst}}$, so the inseparability statement is about the co-frequency regime — $|\Delta f|T_{\text{burst}} \lesssim O(1)$ with comparable powers and short bursts — not about the source count as such. Second, the identifiability is of the effective coefficients, not of the carrier phases in isolation (Proposition 2). The proposition also legitimises the coupling-matrix formulation of Section IV-D: the joint discrete structure is what makes the per-source gains estimable at all.

D. Frequency-offset CRB: why sync goes after separation

For a single tone $x_n = Ae^{j(2\pi f_n/f_s + \phi)}$ in complex AWGN with unknown phase, the Rife–Boorstyn bound [20] on any unbiased frequency estimate is

$$\text{var}(\hat{f}) \geq \frac{6f_s^2}{(2\pi)^2 \eta N(N^2 - 1)}, \quad \eta = A^2/\sigma^2, \quad (8)$$

which for our burst ($N = 4096$, $f_s = 16$ kHz) evaluates to a standard deviation of 0.042 Hz at $\eta = -5$ dB and 0.0024 Hz at 20 dB. For two equal-power superimposed tones we compute the Fisher information exactly, $J_{ij} = (2/\sigma^2)\text{Re} \sum_n (\partial s_n/\partial \theta_i)^* (\partial s_n/\partial \theta_j)$, and invert it numerically, under two nuisance models: $(f_1, f_2, \phi_1, \phi_2)$ with the amplitudes known, and $(f_1, f_2, \phi_1, \phi_2, A_1, A_2)$ with the per-tone gains unknown — the benchmark's actual condition, since w_k and the fading taps are unknown to the receiver. We stress that either way this is a pure-tone surrogate: the FIM of the unmodulated carrier lines, not of the full communication waveform (whose data symbols, pulse shaping and fading a complete model would carry); it bounds the frequency information that the carrier structure alone provides, which is exactly the information a non-data-aided synchroniser exploits.

Proposition 3 (Two-tone CRB inflation, carrier-only model). With $|\Delta f| = |f_1 - f_2|$, the exact per-tone frequency CRB of the two-tone surrogate inflates over (8) by a factor that diverges as $\Delta f \rightarrow 0$ and decays to unity beyond a few times $1/T_{\text{burst}}$. Under the benchmark's offset law each source's residual offset is $\delta_k = u_k + \epsilon_k$ i.i.d., a trapezoid on $[-5, 10]$ Hz, so $\Delta f = \delta_1 - \delta_2$ ranges over $[-15, 15]$ Hz with the piecewise-quadratic density derived in Appendix A ($\mathbb{E}|\Delta f| = 89/24 \approx 3.71$ Hz, median 3.23 Hz, 90th percentile 7.56 Hz, and $P(|\Delta f| < 1/T_{\text{burst}}) = 0.587$). Averaging the exact inflation over this law gives median $1.70\times$ and 90th percentile $7.6\times$ with phases the only nuisance parameters, and median $4.6\times$, 90th percentile $287\times$ when the gains are also unknown (Fig. 3, left).

Remark 1 (From the carrier-only surrogate to the communication waveform). The pure-tone FIM models the information a non-data-aided synchroniser can exploit: the spectral line of the carrier. To locate what the modulation structure adds, we computed the Fisher matrix of the full benchmark waveform under the minimal data-aided model — known symbols, modulation and pulse shaping; unknown carrier offsets and unknown 3-tap complex channels (14 real parameters, phases absorbed into the free channel taps, exactly the entanglement of Proposition 2) — by exact differentiation of the noiseless mean signal. The two-source frequency CRB shows no coupling: the

median inflation over symbol/channel realisations stays at $1.01 \times$ for every $|\Delta f| \in [0, 16]$ Hz (90th percentile $\leq 1.02 \times$; modulation-agnostic over the benchmark pairs), against the pure-tone divergence at the same offsets (Fig. 3, left). The disambiguating information at co-frequency therefore lives in the data, not in the carrier lines: a non-data-aided receiver cannot reach it, while a fully data-aided one would not need joint processing for the frequencies themselves. Our joint loop sits in between — its hard decisions act as semi-pilots, which is precisely why it beats per-slot spectral-line synchronisation, and why its residual gap to the oracle-frequency reference (Section V-D) is the price of decision errors rather than of frequency coupling. The surrogate is thus a diagnostic of the non-data-aided route, not a bound on the communication receiver.

Validation and framing. Proposition 3 is an architectural statement about the non-data-aided route: estimating two carrier frequencies from the mixture by their spectral lines is information-poor precisely in the co-frequency regime. The experiments test the two ways out — per-slot sync after separation, and joint sync-detection on the mixture — and reject the first (E3, Section V-C) in favour of the second (E4); Remark 1 explains why the second can work at all. Throughout, the CRB is used as a benchmark for unbiased estimation, not as a measure of algorithmic efficiency: measured errors decompose as $\text{MSE} = \text{Var} + \text{Bias}^2$, and wherever acquisition bias or fading-induced ISI dominates — the high-SNR floors below, and the interferer-capture bias at $K \geq 2$ (Section V-C) — the gap to the CRB is squared bias, which no unbiasedness assumption prices. Fig. 1 closes the loop on the two-source case the proposition addresses: a spectral-line two-tone estimator tracks the exact CRB within $1.0\text{--}1.3 \times$ in 15 of 18 cells (worst $2.4 \times$) once $|\Delta f| \gtrsim 1.6/T_{\text{burst}}$, and below $\approx 1/T_{\text{burst}}$ undergoes the classical capture effect [28] (SNR-flat RMSE saturation, $\pm 0.6\text{--}0.7$ Hz per-tone bias, 89–98% failure at 1 Hz) — bias-dominated, as the inflation predicts. The right panel of Fig. 3 overlays the measured frequency errors of our BlindCarrierSync on $K=1$ bursts (100 per modulation and SNR): BPSK/QPSK run within $1.5 \times$ of the CRB standard deviation at $-5\text{--}0$ dB ($\leq 1\%$ failures, ≤ 0.013 Hz bias); the $M=8$ line of 8PSK sinks under the noise floor at -5 dB (84% failure) and weak-line bursts cost $\leq 10\%$ failures for 8PSK/16QAM at mid SNRs (Section IV-B); and at 20 dB every modulation’s inlier RMSE flattens to an SNR-independent $0.01\text{--}0.07$ Hz floor set by fading-induced ISI, not noise (up to $23 \times$ the CRB for 16QAM; the 8PSK all-burst RMSE is 0.135 Hz, carried by 1% outliers).

IV. Method: the joint synchroniser-detector

Section III designates the receiver: carriers, gains and symbols must be estimated jointly, on the mixture. We present that headline receiver first (Section IV-A) — it is the arm that produces the paper’s headline number — then the $K=1$ branch (BlindCarrierSync, Section IV-B), and finally the diagnosed baseline chain separate $\rightarrow$ per-slot sync $\rightarrow$ joint re-detection (RS-SlotSepNet, Sections IV-C–IV-D) that E3 and E4 reject as a detection front-end.

A. The headline receiver: ECM joint synchronisation-detection on the raw mixture

The receiver operates directly on the symbol-rate mixture $r_{1:N}$ of (2): no separation, and no oracle carriers, phases or symbols; the two modulations are the known semi-blind input. It maximises the joint likelihood over the discrete symbol pair and the continuous

carrier/gain parameters by an expectation-conditional-maximisation (ECM) [26] scheme: joint hard decisions on the $\leq 16 \times 16$ hypothesis grid (3) alternate with closed-form complex least-squares refits of the 2×2 coupling matrix A (classification-EM), nested in a coarse-to-fine two-dimensional frequency search — a 66-point coarse grid over $[-5, 10]^2$ Hz, a 0.2 Hz fine grid around the top-4 candidates, a full-rate convergence, and a 7×7 NLS polish — with sixteen gain-phase restarts covering one symmetry sector of Proposition 2 against swapped or degenerate minima; multi-start solutions are ranked by the residual energy $\mathcal{E}$. In this memoryless model every step is an exact coordinate minimiser of $\mathcal{E}$, so each round is energy-monotone by construction; Appendix B gives the full procedure and convergence criteria.

The ISI-aware extension (V4) replaces the memoryless observation by a per-source $L=5$ -tap centred symbol-rate response $r_n = \sum_k e^{j2\pi\Delta f_k n T_s} \sum_{l=0}^{L-1} g_k[l] c_{k, n-(l-\lfloor L/2 \rfloor)}$ (centred matches the generator’s zero-delay fading; $L=7$ saturates), warm-started from V1: joint grid decisions on a parallel-iterated ISI-cancelled stream alternate with a closed-form $2L$ -tap least-squares refit and a tap-aware NLS polish. The tap refit is an exact conditional minimiser but the ISI-cancellation decision step is not (the taps couple symbols across time), so V4 is a monotone coordinate-refinement in the generalised-EM sense, enforced by an explicit fallback to the V1 energy (never triggered); V1 is V4 at $L=1$, doubling as the memoryless ablation. The pair costs ≈ 1.6 s per burst on one CPU core — an offline proof of concept, not a real-time stage.

B. BlindCarrierSync: the $K=1$ branch

Each occupied slot’s passband burst $\hat{s}(t)$ is synchronised blindly, with no learning and no oracle input. The residual carrier offset after down-conversion at the nominal f_0 lies in $[-5, +10]$ Hz. The final algorithm — reached by measurement, and simpler than the textbook pipeline in two places — is:

- 1) Coarse down-conversion at f_0 .
- 2) Sample-level coarse frequency. Apply the RRC matched filter first (a 10.7 dB noise-bandwidth reduction that is essential: on the raw burst, transition-slew spurs in the nonlinearity spectrum exceed the carrier line for 8PSK/16QAM even at high SNR), then form the M -th power (M the rotational symmetry order; phase-only normalisation for PSK, amplitude-weighted powers for 16QAM [19]). The line at $M\Delta f$ is located by an $8 \times$ zero-padded FFT with parabolic peak interpolation.
- 3) Symbol-level refinement and wrong-peak rescue. Three candidates: the coarse estimate, a narrow (± 2 Hz) re-estimate around it, and a full-window (± 13 Hz) re-estimate, all from the M -th-power line search on the decimated symbol stream (256 clean constellation samples instead of 4096 slew-corrupted ones). The candidate maximising the product of two decision-free coherence scores — sample-level line strength and symbol-level $|\text{mean}(z^M)|$ — wins; the product works because the two scores have disjoint failure modes (deep-fade bursts keep a strong sample-level line while symbol-level coherence is fooled by ISI, and vice versa).
- 4) Demodulation with the winning estimate: correction, matched filter, symbol grid, unit-power normalisation.
- 5) No decision-directed phase refinement. The classical DD line fit of the decision-referred phase residual is fold-biased under fading: wrong decisions fold

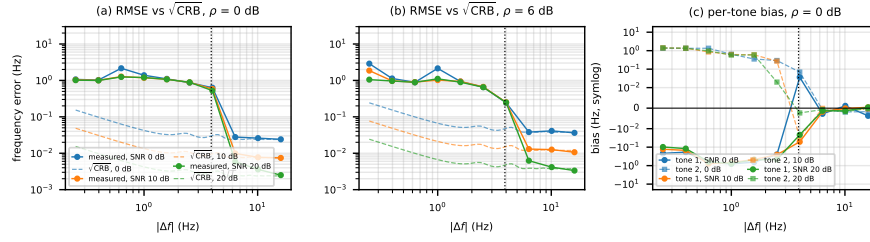


Fig. 1. Direct validation of Proposition 3 ($K=2$ carrier-only model): measured spectral-line two-tone estimation vs. the exact two-tone CRB. (a),(b) pooled RMSE (solid) vs. $\sqrt{\text{CRB}}$ (dashed) at amplitude ratio 0 dB / 6 dB; (c) per-tone bias (symlog, 0 dB); vertical line: $1/T_{\text{burst}}$.

the measured phases toward the nearest grid point, biasing the fitted intercept ($\approx 6^\circ$) and increasing SER at high SNR (8PSK at 20 dB: $0.028 \rightarrow 0.087$). Non-data-aided alternatives (Viterbi–Viterbi symbol phase, min-distance phase scan) are no better under the 3-tap channel. The residual constant phase is small after frequency correction and is absorbed by the M -fold ambiguity resolution of the scoring (differential coding in a deployed system).

Known boundaries, analysed. The M -th-power line amplitude at the true residual factorises as $|\mathbb{E}[c^M]| \cdot |\sum_l h_l^M|$ with h_l the fading taps (the cross-moments vanish under the alphabets’ rotational symmetry). For PSK, c^M is a deterministic constant — $|\mathbb{E}[c^M]| = 1$, zero data self-noise. For 16QAM, $\mathbb{E}[c^4] = -0.68$: the three amplitude rings contribute opposite-signed fourth powers, so the coherent line is 3.3 dB weaker and rides on per-symbol self-noise ($\text{Var}(c^4) = 2.66$). Under the benchmark’s 3-tap Rayleigh channel the second factor has mass near zero — $P(|\sum_l h_l^4| < 0.15) \approx 7\%$ — and for those bursts no fourth-power method has a usable line (the measured symbol-level line strength tracks $|\mathbb{E}[c^4] \sum_l h_l^4|$ with pooled correlation 0.75 across modulations; within a single modulation the channel-tap factor alone correlates weakly at 20 dB, 0.03–0.19). At $\text{SNR} \leq -5$ dB the $M=8$ line of 8PSK sinks under the noise floor. Both are intrinsic to short-burst non-data-aided estimation, and both are reported rather than engineered away (Section V-A).

C. The diagnosed baseline I: slot separation (SlotSepNet)

The separator is our own slot-based design, used here as a fixed, pretrained baseline — it is not the contribution, and E4 will show it should not be the detection front-end. A complex-valued convolutional encoder (complex arithmetic via paired real operations [22]) feeds four complex residual blocks with complex squeeze-and-excitation attention [21]; the representation is read out by $K_{\text{max}}=4$ mask slots, each producing a separated waveform, with a shared occupancy head scoring slot activity and a count head aggregating the occupancies ($\approx 254K$ parameters). Training uses the benchmark’s mixtures ($K \in \{1, 2, 3\}$; 2000 mixtures/epoch, batch 16, Adam 10^{-3} , 100 epochs) with a variable-count permutation-invariant loss [11] on SI-SDR [23] plus occupancy cross-entropy and an MSE anchor; five seeds (42–46) give the checkpoint population E3/E4 average over. Its quality is measured, not asserted: on the test grid the separator reaches SI-SDR improvement 9.6/3.2/2.8 dB at $K=1/2/3$ with slot occupancy recall 0.98/0.98/1.00 (precision 0.89/0.68/1.00 — the $K=2$ precision is the surplus-slot behaviour of Section V-C), and two cross-family checks on the same $K=2$ cells: the C-SE convolutional separator of our companion work [27], evaluated zero-shot, reaches SI-SDRi 3.5 dB, and the

published CNSE separator [1] (6.7M parameters, $26 \times$ ours, retrained on the benchmark’s $K=2$ mixtures, five seeds) reaches 6.0 dB — so the E3 premise does not hinge on one architecture, and Section V-D prices what the stronger separator buys the waveform route. The waveform model’s job is only to hand the receiver per-slot streams whose residual is structured leakage — how good those streams are for detection is an empirical question, answered (negatively) by E4’s V2 arm.

D. The diagnosed baseline II: JointPairDetector ($K=2$)

Given two synchronised unit-power slot streams $z_{1,n}, z_{2,n}$ and their modulation hypotheses, the joint log-likelihoods over the $\leq 16 \times 16$ hypothesis grid are

$$\ell_n(i, j) = -\|z_n - A[c_i; c_j]\|^2 / \sigma^2, \quad z_n = [z_{1,n}; z_{2,n}], \quad (9)$$

with $A \in \mathbb{C}^{2 \times 2}$ the coupling matrix whose columns absorb the per-source complex gains (Proposition 2). A is estimated blind by a classification-EM (hard-EM) procedure from $A = I$: the E-step makes the joint hard decision (9) under the current A ; the M-step is the closed-form complex least-squares refit $A \leftarrow ZC^H(CC^H)^{-1}$ on the current decisions. Both steps are exact coordinate optimisations of one least-squares objective, so the objective is non-increasing per round (the loop stops at a 10^{-5} relative decrease or 30 rounds). We call the scheme by its proper name: classification-EM rather than soft EM with posterior responsibilities, and — nested inside the frequency search of Section IV-A — an ECM scheme over the discrete symbols and the continuous $(A, \Delta f_1, \Delta f_2)$ jointly. Because the landscape at $\text{SIR} \approx 0$ dB can land in swapped or degenerate minima, we restart from a small grid over off-diagonal magnitude and phase (the known operating range $|a_{ij}|/|a_{ii}| \lesssim 1.5$ from $w_k \sim \mathcal{U}(0.4, 0.6)$) and keep the minimum-energy (maximum-likelihood) solution (same-alphabet pairs retain a near-degeneracy — Section VII; E4 measures the effect end to end).

We ran this control: the baseline separator fine-tuned with an additional differentiable soft-demodulation term (soft constellation cross-entropy on the PIT-assigned pairs of the baseline loss, $\lambda_{\text{ser}} \in \{0.1, 1.0\}$, five seeds, the identical data stream) moves the compensated SER of the E4 waveform route by at most 0.12 points ($0.5337 \rightarrow 0.5334/0.5325$; BER and the 3.2 dB SI-SDR improvement unchanged within seed noise) — the tested loss-level modification does not remove the floor of Theorem 2 on this benchmark: within the tested waveform-separator objective family, the remedy remains structural. (Architecture-level separator strength is a different lever: E4’s CNSE cross-check, Section V-D.)

V. Experiments

The benchmark is the variable- K co-frequency test grid of Section III-A (deterministic test seed 99999; SNR

TABLE I

E1: compensated SER per modulation, oracle vs. blind carrier synchronisation ($K=1$; 200 bursts per cell, two test grids; blind = BlindCarrierSync, genie M -fold resolution). The nominal no-sync proxy sits at 0.64–0.70 across SNR.

SNR (dB)	BPSK		QPSK		8PSK		16QAM	
	or.	blind	or.	blind	or.	blind	or.	blind
−10	0.161	0.159	0.308	0.502	0.526	0.796	0.699	0.847
−5	0.091	0.092	0.171	0.177	0.331	0.756	0.510	0.584
0	0.057	0.059	0.083	0.086	0.179	0.212	0.304	0.321
5	0.029	0.030	0.049	0.053	0.101	0.124	0.164	0.189
10	0.012	0.011	0.028	0.030	0.066	0.088	0.094	0.114
15	0.009	0.009	0.021	0.026	0.032	0.060	0.052	0.080
20	0.007	0.007	0.009	0.012	0.018	0.054	0.027	0.065

grid $\{-10, \dots, 20\}$ dB). The evaluation truth is an offset-compensated receiver — true-carrier down-conversion plus phase alignment — which serves as the idealised limit any synchroniser is scored against. The operating envelope throughout is $K \in \{1, 2\}$ sources with known modulations — E5 measures both boundaries and finds them structural. A note on terminology: “blind” here qualifies the estimate path — no oracle carriers, phases, symbols or modulations are used by any estimator, and the receiver stages are training-free — not the operating assumptions, which are a known nominal carrier and offset range, the benchmark’s fixed timing grid, and per-source modulation knowledge. The receiver is therefore precisely a semi-blind, modulation-conditioned one. E1–E6 below are complete: E1–E2 are CPU and deterministic; E3 ran on the GPU server over the five baseline separator checkpoints; E4’s mixture-route variants (V1/V3/V4) are training-free and deterministic, and its slot-route variants (V0/V2) share E3’s five checkpoints; E5–E6 are CPU and deterministic.

A. E1: does blind sync approach oracle sync?

On $K=1$ test cells (no separation, isolating the synchroniser; 200 bursts per modulation–SNR cell, two grids) three receivers demodulate the same bursts against the same reference labels: the nominal proxy (down-conversion at f_0 , no synchronisation), the oracle compensated receiver, and BlindCarrierSync with the true modulation hypothesis, scored with genie resolution of the M -fold phase ambiguity (Proposition 2; the deployment answer, differential coding, is measured in Section VII). Table I reports the per-modulation SER.

Blind synchronisation meets its design target — oracle-level SER — for BPSK and QPSK at $\text{SNR} \geq -5$ dB (gaps ≤ 0.6 pts; BPSK’s small inversions are reference-label noise). The two honest exceptions are larger than a low-count pilot suggested: 8PSK runs 2–4 points above the oracle at $\text{SNR} \geq 0$ dB — stable across the two independent grids — and 16QAM runs +2–4 points above it at $\text{SNR} \geq 5$ dB (+1.7 at 0 dB; weak-line bursts and residual-phase sensitivity; Section IV-B); and the M -th-power line collapses below the noise floor for 8PSK at $\text{SNR} \leq -5$ dB and for QPSK at -10 dB — a physical limit of non-data-aided estimation at 256-symbol bursts, not of this particular estimator (where the line survives, the frequency error is within $1.5\times$ of the CRB; Fig. 3).

B. E2: theory validation

Fig. 3–4 collect the measurement validation of Section III; Fig. 2 shows the Monte-Carlo curves behind Section III-B directly. The bounds sandwich the measured operating points: the union bound sits above the joint Monte-Carlo SER at every phase grid point (asserted computationally), the exact evaluation (6) sits on the separate Monte-Carlo curve (within 0.0036) and converges

TABLE II

E3: blind-minus-oracle compensated SER (percentage points, 5-seed mean), per K and SNR. Pooled, the blind arm loses $+8.5 \pm 0.1$ pts (population std; 5/5 seeds positive); pooled BER degrades $0.2651 \rightarrow 0.3519$.

K	ΔSER (pts) at SNR (dB)						
	−10	−5	0	5	10	15	20
1	+14.7	+16.9	+2.3	+2.0	+1.3	+0.7	+1.8
2	+9.7	+10.9	+9.4	+8.1	+8.3	+7.6	+8.1
3	+7.4	+7.9	+9.6	+9.8	+9.7	+10.7	+10.1

to the continuous-ensemble floor (7) (0.4999 resp. 0.8380), the marginal-MAP Monte-Carlo removes the joint rule’s low-SNR reversal, and the measured synchroniser errors sit above the CRB with the expected inefficiency factor. The 16QAM column of the Monte-Carlo validation is the quantitative reason the end-to-end gain should be expected on PSK pairs rather than QAM pairs.

C. E3: end-to-end blind pipeline

E3 scores the end-to-end pipeline under two receivers sharing one scoring code. The baseline separator of Section IV-C (five seeds, no fine-tuning) separates each test mixture; the occupied slots (occupancy > 0.5 , Hungarian-matched to the true sources on SI-SDR, matched pairs only) are demodulated by the oracle compensated receiver (true-carrier down-conversion plus phase alignment) or by BlindCarrierSync, scored by compensated SER/BER with genie M -fold rotation resolution ($K \in \{1, 2, 3\}$, 100 mixtures per $K \times \text{SNR}$ cell, test seed 99999).

The oracle arm is already this paper’s first empirical result — the waveform-to-bit gap, measured directly. With separation in place and synchronisation idealised away, the pooled SER is 0.5027 (per-seed range 0.5019–0.5037; BER 0.2651), but the per- K split shows where that number comes from: 0.1454 at $K=1$ against 0.5534/0.5858 at $K=2/3$. Theorem 2 pins down the $K=2$ number per constellation: the oracle arm pools 0.454 on QPSK sources (exact $\text{SIR}=0$ dB expression averaged over the same SNR grid: 0.467) and 0.821 on 16QAM sources (floor 0.838), while BPSK sources, whose floor is 0, pool 0.251 on noise alone; the pooled 0.5534 is the modulation-weighted mixture of these levels, all far above the $K=1$ level: the separator works as a waveform model — its slots match the true sources closely enough to be Hungarian-paired on SI-SDR — yet through any separate-then-detect receiver that quality does not reach the bits. Replacing the oracle receiver by BlindCarrierSync makes this strictly worse, and because both arms share the scoring code, every gap below is attributable to synchronisation alone.

Table II decomposes the penalty, and the decomposition is the result. At $K=1$ the blind pipeline behaves exactly as E1 predicts: the gap is $+0.7$ – $+2.3$ pts at $\text{SNR} \geq 0$ dB and blows up only at -10 – -5 dB, where the M -th-power line sinks below the noise floor (Section V-A). At $K \geq 2$ the gap is instead SNR-flat: $+7.6$ – $+10.9$ pts across the full SNR range. The mechanism is visible in the synchroniser’s own error statistics: the median blind frequency error on $K \geq 2$ slots is 2–4 Hz at every SNR (against 0.007–0.04 Hz on $K=1$ slots at $\text{SNR} \geq 0$ dB). The residual interferer leaking into each slot carries its own M -th-power spectral line, and that line pulls the per-slot carrier estimate off the true carrier — a structured, SNR-independent bias, not noise. Two controls pin the mechanism down. First, a crude successive-interference-cancellation variant (sync each slot after subtracting the other matched slots from the mixture) does not help: pooled $K=2$ SER 0.6484 vs. 0.6422 for plain blind sync — the subtracted estimates are

TABLE III

E4: per-source SER on the $K=2$ test cells. V0/V2 average over the five separator checkpoints on the reference grid (test seed 99999); V1/V3/V4 are deterministic per grid and average over five independent test grids (seeds 99999/88888/77777/55555/31337; 95% seed-level CIs $\pm 0.004/0.004/0.005$ for V1/V4/V3); marginal-MAP re-decision changes nothing (V1 0.5388 vs. 0.5392; V3 0.4102 vs. 0.4106). The reference rows are the waveform route (separate, oracle sync), symbol-wise and with per-slot PSP/Viterbi (5-checkpoint means), and the same PSP route with the separator replaced by the retrained published CNSE [1]; blind demodulation of the raw mixture scores 0.6427. The last column is the pooled Gray-mapped BER (same decisions).

Variant	SER at SNR (dB)							pooled	BER
	-10	-5	0	5	10	15	20		
V0 slots $\rightarrow$ blind sync $\rightarrow$ EM-A	0.674	0.628	0.634	0.559	0.590	0.590	0.580	0.6078	0.353
V1 ECM on raw mixture	0.688	0.628	0.554	0.518	0.474	0.458	0.455	0.5392	0.312
V2 ECM on slot streams	0.702	0.660	0.634	0.547	0.531	0.550	0.548	0.5960	0.354
V4 ISI-aware ECM ($L=5$)	0.689	0.628	0.551	0.510	0.461	0.441	0.438	0.5310	0.306
V3 oracle-frequency ECM	0.608	0.522	0.422	0.374	0.331	0.315	0.303	0.4106	0.215
waveform route, oracle sync, symbol-wise	0.592	0.555	0.564	0.526	0.544	0.548	0.540	0.5527	0.285
waveform route, oracle sync, PSP/Viterbi	0.585	0.537	0.545	0.493	0.526	0.527	0.522	0.5337	0.266
waveform route, CNSE separator, oracle sync, PSP	0.580	0.508	0.488	0.372	0.410	0.378	0.373	0.4440	0.213

Statistical reporting. The unit of inference throughout E4 is the mixture burst: each of the five independent test grids holds 7 SNR values $\times$ 100 bursts = 700 bursts, and pooled SERs aggregate all bursts of a route (95% CIs over the five grids). Paired contrasts use the paired per-burst difference of two routes on the same bursts: the V1–V4 contrast pools all 3500 paired bursts of the five grids, while the headline waveform-route contrast pairs the 700 bursts of the reference grid (seed 99999), the route arm averaged over its five separator checkpoints per burst. For every paired contrast we report a paired-bootstrap 95% CI of the mean per-burst difference together with a Wilcoxon signed-rank p -value; the two target different estimands — the mean gap versus symmetry of the paired differences about zero — so a bootstrap interval containing zero can coexist with a significant Wilcoxon p , as for the headline contrast below. A tie is claimed only when the paired interval contains zero. The route arm’s five checkpoints were evaluated on the reference grid only (each additional grid costs a PSP/Viterbi pass per checkpoint), so the headline contrast is paired on that grid’s 700 bursts while V4’s own uncertainty is the five-grid CI above. The paired differences are diffuse rather than concentrated: the V1–V4 contrast is positive on 59% of its 3500 bursts (per-grid 0.57–0.60) and the headline route contrast on 36% of its 700 — a small mean shift spread across many bursts, not a subset effect.

Results. Table III orders the variants as sanity demands, $V3 < V4 < V1 < V2 < V0$, and delivers three findings. First, the headline, which the paired tests make precise. The training-free joint loop on the raw mixture ties the strongest waveform route at the paper’s separator quality: pooled SER 0.5310 (V4, 95% CI [0.5261, 0.5360] over five independent test grids) against 0.5337 for the oracle-received waveform route upgraded with per-slot PSP/Viterbi sequence detection [24], [25] (decision-directed $L=3$ channel estimation; 0.5527 for its symbol-wise special case) — the route-minus-V4 paired per-burst difference is $+0.003$ with 95% CI $[-0.011, +0.018]$ (Wilcoxon $p=4.8 \times 10^{-4}$; $n=700$ paired bursts) — while using no oracle carriers, phases or symbols during inference, against 0.6427 for blind demodulation of the unseparated mixture (−11.2pts). The contrast is SNR-structured, and the structure is the content: at $\text{SNR} \leq 0\text{dB}$ the oracle-assisted separate routes win by 4–11 points (their advantage is the oracle synchroniser, not the architecture); at $\text{SNR}=5\text{dB}$ the joint loop ties the PSP arm (+0.035, CI $[-0.000, +0.070]$, Wilcoxon $p=0.19$) and beats only the symbol-wise arm (+0.068, CI $[+0.029, +0.106]$, $p=0.006$); from 10dB upward it wins the PSP arm by +0.07–+0.08 per cell. (Table III’s V1/V3/V4 cells average five grids

while the route rows use the reference grid; all win/tie statements are same-grid paired contrasts, not row-wise differences.) The memoryless special case V1 already ties the PSP route (0.5392, CI [0.5342, 0.5441]). The bit-level view (pooled Gray-mapped BER, Table III) is more nuanced: V4’s 0.306 trails the PSP route’s 0.266 pooled — the 16QAM-involving pairs carry four bits per symbol and no route recovers them — while on PSK-only pairs V4 leads (0.236 vs. 0.250), mirroring the SER structure. We also score the PSK-pair bursts with a rotation-invariant, differential-equivalent label ($d_n = \hat{c}_n \hat{c}_{n-1}^*$ quantised to the M -th roots; $n=383$ PSK-pair bursts of the reference grid for V1/V4, the route arms averaged over five checkpoints, 1915 evaluations) that removes the scoring genie entirely. It measures the change rate of the per-symbol phase error — a phase-slip rate, not a differential-receiver SER — rewarding slowly-varying phase errors regardless of magnitude, so it is not on the same axis as the genie-resolved SERs above; we report it only as a secondary, rotation-free view: V4 scores 0.372/0.244 (SER/BER equivalents) against the PSP route’s 0.537/0.357 — the joint loop’s residual phase error is mostly slow acquisition drift, the waveform route’s noise-driven. Two caveats: the data are not differentially encoded, so absolute bits remain rotation-ambiguous under either score; and genuine differential detection on clean streams costs +0.03–+0.10 SER at 20dB, paid identically by oracle and blind arms (Section VII). The architecture contrast, isolated from acquisition by the oracle-frequency arm, is larger: V3 beats the oracle-received PSP route by 12 pooled points (0.4106 vs. 0.5337), growing to +22 points at 20dB — joint detection, not synchronisation, is where the headroom lives.

Cross-architecture check (CNSE). Is the waveform route’s ceiling set by our separator rather than the architecture? We retrained the published CNSE separator [1] (6.7M parameters, $26\times$ ours, $K=2$ -specialised training; SI-SDRi 6.0dB vs. 3.2dB) and ran it through the identical route (Table III, last row). The route improves by 9 pooled points (0.5337 $\rightarrow$ 0.4440; BER 0.266 $\rightarrow$ 0.213): it is separator-quality-limited, and Theorem 2’s floor is confirmed as a property of the linear-residual regime it models, not of separation as such — consistent with the true-stream control (0.042). (i) At matched oracle frequencies the joint loop still wins: V3 0.4106 vs. 0.4440. (ii) The CNSE route’s advantage is entirely the oracle synchroniser: replacing it by per-slot blind sync costs +9.2 pooled points (0.5356) — inside E3’s +7–11 capture band, with 36–45% of bursts still acquiring more than 1Hz off the true offset at $\text{SNR} \geq 5\text{dB}$ despite the 6dB cleaner slots (the signature of interferer-line capture)

— so blind-versus-blind the joint loop again ties the strongest waveform route (0.5310 vs. 0.5356). (iii) Under the rotation-invariant differential-equivalent score (PSK pairs; the phase-slip-rate estimand above) the ordering persists: V4 0.372 vs. CNSE oracle-PSP 0.438 and CNSE blind-PSP 0.442. Second, V0 is worse than the waveform route it augments (0.6078 vs. 0.5527): chaining the biased per-slot sync of E3 into the joint detector poisons it, confirming that the E3 failure is in the sync stage, not the detector. Third, V2 — the slot-aided architecture — loses to V1 by +5.7 pts, positive on all 5/5 seeds (+4.3 pts against the waveform-route reference, also 5/5). Direct measurement on 40 held-out $K=2$ bursts at 20 dB explains why: the mask-separated slot streams carry a median per-burst ratio of 1.6–1.84 $\times$ the mixture’s converged-ECM residual energy per channel-symbol (PSK pairs: median 0.0274 vs. 0.0122; 16QAM-involving: 0.0297 vs. 0.0148) — mask separation distorts the linear two-source superposition (1) that the ECM likelihood assumes, and the distortion costs more than the leakage suppression buys (within-burst phase drift is 1.3–1.6 $\times$ larger on the slots, magnitude drift only 1.0–1.3 $\times$). The positive control is the same ECM loop on the two true source streams: SER 0.042 vs. 0.130 for mixture-ECM on the same bursts — the joint loop would win decisively if the separation were clean. At the current separator quality it is not, and the operating answer on this benchmark is: joint detection should run on the mixture; separation serves slot generation and localisation, not detection (its learned count head is unreliable at $K=2$; Section V-C). The remaining V4–V3 gap (0.531 vs. 0.411) is the frequency-acquisition residual, the price of Proposition 3’s two-tone inflation paid once, jointly, instead of twice, per slot.

Where the gain lives. Splitting the pool by alphabet pair: on PSK-only pairs V4 reaches 0.378 (V1: 0.390; V3: 0.254), while 16QAM-involving pairs sit at 0.726, essentially unchanged from V1’s 0.729 (V3: 0.610) — exactly the concentration E2’s $d_{\min}(\varphi)$ analysis predicted (Section V-B): for 16QAM-involving pairs the dense sum constellation leaves no headroom at the burst SNRs, and neither joint detection nor ISI modelling can change that. The pooled figure therefore understates the joint loop’s gain on constant-modulus constellations.

The SNR curve is non-monotone (V1 best at 10 dB, rising again toward 20 dB): below the knee the loop is noise-limited; above it, the memoryless two-source model (2) itself is the floor. V4 confirms the attribution with a signature rather than a single number: the share of the V1→V3 gap that V4 closes on PSK-only pairs grows from 1% at –10 dB to 20% at 20 dB on the reference grid (3/7/15/14% in between; the five-grid average is non-monotone, so this is a single-grid signature) — an ISI floor must vanish when noise dominates and dominate when noise vanishes, and it does; the 20 dB PSK floor itself moves 0.2408 → 0.2169. A reference check with oracle frequencies on 20 dB QPSK pairs isolates the same budget: the $L=5$ loop reaches 0.062 vs. 0.107 for the memoryless oracle (true-stream control: 0.042). V4’s pooled margin over V1 (0.5310 vs. 0.5392) is small precisely because the budget it attacks only opens at high SNR and on PSK pairs — but it is statistically solid (paired per-burst CI [+0.0074, +0.0089], $p \approx 10^{-90}$, $n=3500$ paired bursts) and it costs nothing elsewhere (energy-monotone, zero fallbacks, Table III).

Implementation remarks. The EM inside the frequency search must run to convergence at every ranked candidate (truncated, the residual-energy ranking is inverted); every frequency move must be re-evaluated with EM re-converged (a decision-frozen NLS cannot leave a bad-

decision basin); the gain-phase initialisation must cover one symmetry sector finely. Measured and rejected: M -th-power extreme-lines frequency estimation (cross terms / interferer power dominate at SIR ≈ 0 dB) and decision-directed frequency refinement (decision flips create false spectral lines).

E. E5: blind modulation, $K=3$ probe, and ablations

E5 measures the two boundaries of the operating envelope — blind modulation handling and $K \geq 3$ — and ablates the joint loop’s moving parts. Both boundary measurements are negative, and both are measured, not assumed.

Blind modulation-pair selection (V4B). Everything reported so far conditions on the true modulation pair. Lifting that assumption means choosing among the 10 unordered pairs, each arm costing a full ECM fit. The implemented selectors fail in characteristic ways: the raw converged energy collapses onto the densest hypothesis, picking (16QAM, 16QAM) on essentially every burst — the overfit lives in the per-symbol minimum over a dense grid — and the per-source lock-score two-stage inherits E1’s density bias; logged variants behave the same (held-out even/odd energy identically, unnormalised soft evidence still densest) except per-hypothesis- σ^2 evidence, which overcorrects into always-(BPSK, BPSK). The one working selector is normalised soft evidence: log-sum-exp over the hypothesis grid minus the term-count penalty $\log(M_1 M_2)$, with a shared σ^2 from the best fit. Even with it, the joint classify-and-detect pipeline (wrong pair $\Rightarrow$ SER 1) reaches pair accuracy of only 0.11/0.53/0.40 at SNR 0/10/20 dB (chance 0.10) and end-to-end SER 0.987/0.610/0.762, pooled 0.786 against 0.488 for known-modulation V4 on the same cells. The per-pair breakdown is the diagnostic: at 10 dB the BPSK/QPSK-only pairs are identified reliably (0.86–0.88), 8PSK-involving PSK pairs reach only 0.36–0.50 (six-class PSK pool 0.65), while 16QAM-involving pairs sit at 0.11–0.78; at 20 dB the selector drifts toward 16QAM-involving hypotheses — the density bias of the evidence measure reasserting itself where the noise no longer masks it. Conditional on a correct pair decision the detector is intact (0.880/0.264/0.404 by SNR). The tested evidence-based selectors are insufficient: what is missing is a receiver stage (a dedicated modulation-recognition front-end), so known modulation remains the operating assumption (Section VII).

$K=3$ feasibility probe. Extending the joint loop to three sources (sequential-extraction initialisation: a $K=2$ ECM fit, subtraction, an M -th-power line search for the third carrier; then a three-source converged phase-grid EM with centred $L=3$ taps and PIT over the six assignments) fails at the acquisition stage we tested: on PSK-only triplets (13 and 8 bursts, 6 and 5 distinct, at SNR 5 and 15 dB) the resulting $K=3$ ECM scores 0.622/0.572 against the same-cells oracle mixture baseline’s 0.516/0.458 — the $K=2$ fit locks a phantom pair, after which the residual’s M -th-power line is meaningless. We state the negative precisely: the tested $K=3$ acquisition strategy — sequential extraction plus M -th-power line search — is unreliable; whether a different acquisition stage (e.g. a joint three-tone grid) makes $K=3$ feasible is open, and no impossibility is claimed. This paper’s scope is $K \in \{1, 2\}$ — $K=1$ by BlindCarrierSync, $K=2$ by V4.

Ablations. On a subset of the $K=2$ grid (SNR {0, 10, 20} dB, 25 cells per SNR, test seed 99999; deterministic) the NLS polish contributes nothing (+0.2 pp), dropping the restart set to the top-1 coarse candidate costs 0.8 pp, and the ISI taps show the expected dose-response, SER falling monotonically from $L=1$ (V1, 0.5202) through

$L=3$ (0.5130) to $L=5$ (0.5077) — the V4 gain is carried by the channel model, not any single implementation choice.

F. E6: robustness sweeps

Three sweeps probe the boundaries of the operating assumptions ($K=2$, QPSK+QPSK, 100 bursts per cell, deterministic; Fig. 5). Carrier separation ($|\Delta f| \in \{0, \dots, 8\}$ Hz): the end-to-end SER shows no co-frequency catastrophe (an 8.3-point U-shaped spread over $|\Delta f|$ at 20 dB, minimum near 4 Hz) — the joint likelihood search substitutes for spectral-line acquisition, exactly the route Remark 1 says the modulation structure enables — while the frequency RMSE stays elevated at co-frequency and the V4–V3 acquisition gap is largest at $\Delta f = 0$ (0.047 at 20 dB), vanishing by 8 Hz, where the ISI-aware model beats the memoryless oracle by 3.5 points. Timing offset ($\tau \in \{0, 0.1, 0.25, 0.5\} T_s$): graceful to $0.1 T_s$ (+0.014–+0.047 SER), collapsing at $0.25 T_s$ (0.58 at high SNR) and near-zero at $0.5 T_s$ — the symbol-rate model assumes the zero-delay grid, so timing recovery is a required deployment stage, and its sensitivity is now measured rather than assumed. Amplitude ratio ($\rho \in \{-10, \dots, 10\}$ dB): graceful within ± 5 dB, weak-source-limited at ± 10 dB (0.43–0.48 at high SNR) — this is also the end-to-end SIR sweep: joint detection is best at $\text{SIR} \approx 0$ dB, the benchmark’s operating point, and the weak stream becomes undecodable as $|\rho|$ grows.

VI. Discussion

Three layers, three statistical objects. The results organise into one statement: waveform-fidelity optimisation, independent synchronisation, and per-source detection optimise different statistical objects. Layer 1 (waveform vs. bits): SI-SDR does not control SER — E3 measures the gap directly. Layer 2 (synchronisation): at co-frequency, independent per-source frequency estimation is information-poor (Proposition 3) and the effective coefficients are identifiable only jointly (Proposition 2). Layer 3 (detection geometry): the joint detector’s SER is governed by $d_{\min}(\varphi)$ of the sum constellation while separate detection has an exact, computable floor (Theorems 1–2). Each layer is closed by a different structure: the third by joint detection on the mixture, the second by the same loop’s decision feedback, the first by nothing that waveform-level training alone provides.

From theory to bits. E3 measures the waveform-to-bit gap directly: a working learned separator, read through an idealised oracle receiver, still tracks the per-constellation separate-detection floor at $K \geq 2$ — the gap is interference-structural, not a matter of synchronisation quality — and the theory makes the structure exactly computable: the floor (Theorem 2) is the analytic form of the measured flatness, and the union bound (Theorem 1) with the measured $d_{\min}(\varphi)$ landscape is the analytic form of the headroom. E3 and E4 then converted the theory into an operating design. The separate-then-sync chain fails at $K \geq 2$ for exactly the reason Proposition 2 names — per slot, the interferer-rotated constellation is a valid hypothesis, so the interferer’s spectral line captures the slot’s carrier estimate (2–4 Hz median bias, SNR-flat) — and the fix is not a better per-slot synchroniser but the joint loop the identifiability analysis points to. That loop, run on the raw mixture, ties the strongest waveform route at the paper’s separator quality with no oracle information during inference, and — the cleanest architecture contrast — beats it by 12 pooled points when both are given oracle frequencies (0.411 vs. 0.534), the gain concentrated on PSK pairs as the $d_{\min}(\varphi)$ landscape

predicts. A retrained published CNSE separator [1] ($26\times$ the parameters, SI-SDRi 6.0 vs. 3.2 dB) does not overturn the contrast: its route reaches 0.444 only through the oracle synchroniser (blind: 0.536) and trails the joint loop at matched oracle frequencies (0.411) and under the rotation-invariant differential-equivalent score (0.438 vs. 0.372; estimand caveat in Section V-D). The slot-aided variant’s defeat is equally informative: it prices mask-separation distortion against leakage suppression, and the true-stream control (SER 0.042) says the price flips sign once separation improves — a concrete target for the next separator.

Two error budgets. The V1→V4→V3 ladder ablates the receiver’s own model (V1 = V4 with $L=1$; V3 = V1 with oracle carriers), decomposing the distance from the headline number to the oracle-frequency reference: the model budget (fading ISI the memoryless likelihood does not span) is worth 0.8 pooled SER points, growing to 20% of the PSK-pair gap at 20 dB; the acquisition budget (blind two-tone frequency estimation under Proposition 3’s inflation) dominates at 12 pooled points and is where any further headroom lives.

VII. Limitations

(i) Scoring resolves the residual M -fold phase ambiguity by a genie; the deployment answer is differential coding [29], measured rather than assumed: on the $K=1$ grid, genuine continuous-stream differential decoding (eval_diff_coding.py) is even more robust than genie-resolved decoding at low SNR (8PSK at -5 dB, 0.458 vs. 0.747) and costs +0.03–+0.10 SER at 20 dB, a penalty the oracle receiver pays equally — the cost is intrinsic to differential detection, not to synchronisation. At $K=2$ the rotation-invariant differential-equivalent score of Section V-D (a phase-slip rate, not a differential-receiver SER) reads $0.381 \rightarrow 0.372$ for V4 on PSK pairs versus $0.440 \rightarrow 0.537$ for the oracle-assisted waveform route; the contrast reflects error structure, and since the data are not differentially encoded, absolute bits remain rotation-ambiguous under either score. The synchroniser fails wholesale for 8PSK below -5 dB SNR; its 16QAM gap of +2–4 SER points at high SNR is dominated by weak-line bursts and residual-phase sensitivity. (ii) Known modulation is an operating assumption of the whole pipeline, not just of E1: E5’s blind modulation-pair classification at $K=2$ reaches 0.11–0.53 accuracy (chance 0.10; 100 cells per SNR) under every selector tried, costing end-to-end SER 0.786 vs. 0.488 known-modulation; conditional on a correct pair the detector itself is intact, so the missing component is a modulation-recognition stage, which this paper does not build. (iii) EM estimation of the coupling matrix is only partially effective for same-alphabet pairs. (iv) The timing grid is fixed by benchmark construction; a deployment version needs timing recovery — the measured sensitivity (Section V-F) is graceful to $0.1 T_s$ and collapses by $0.25 T_s$. (v) The E4 loop models fading ISI only as $L=5$ symbol-rate taps at the fixed zero-delay grid: the SNR curve remains mildly non-monotone (V4 best at 10 dB), the long-tail residual the taps do not span is visible in the oracle-frequency check (0.062 vs. the true-stream 0.042), and 16QAM-involving pairs admit no gain at any SNR. (vi) The ECM loop costs ≈ 1.6 s per burst per V1+V4 evaluation on one CPU core (a 12th-generation Intel i5; the per-candidate cost is linear in N and $M_1 M_2$ and the joint hypothesis set grows as $\prod_k M_k$ — exponential in K , the concrete scaling reason $K=3$ must change acquisition structure, Section V-E). The loop is an offline proof-of-concept receiver, not a real-time one: 1.6 s per burst is $\approx 6\times$ the burst duration

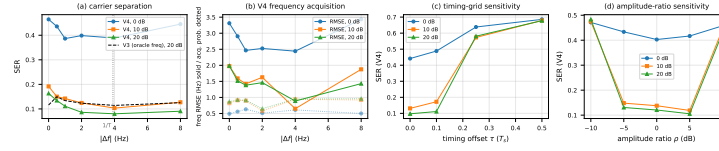


Fig. 5. Robustness sweeps of the V4 joint loop ($K=2$, QPSK+QPSK, 100 bursts per cell; V3 is the oracle-frequency reference). (a) SER vs. carrier separation $|\Delta f|$. (b) V4 frequency RMSE (solid) and acquisition probability (dotted; fraction of bursts whose final frequency error falls below 1 Hz) vs. $|\Delta f|$. (c) SER vs. timing-grid offset τ . (d) SER vs. amplitude ratio $\rho = 20 \log_{10}(w_2/w_1)$.

(0.256 s), the measured peak resident set is ≈ 0.2 GB, and the loop is numpy CPU code, so a GPU accelerates only the separator forward pass, not the joint loop. (vii) V1/V3/V4 are deterministic and training-free, so their evaluation randomness is the test grid itself; we quantify it by re-running E4 on five independent test grids (the seed spread is in Table III), while the V0/V2 variance comes from the separator checkpoints only. (viii) Nothing in the paper decides K : the separator's count head is unreliable at $K=2$ (4.2% exact-count accuracy, Section V-C), so the $K \in \{1, 2\}$ envelope assumes K is known or supplied by an external stage. $K \geq 3$ is out of scope: the sequential frequency acquisition locks a phantom pair on three-source mixtures, and the $K=3$ probe trails the mixture baseline by 10–11 SER points (Section V-E); a genuinely different acquisition stage is needed. (ix) E3's per-pair SERs are conditional on successful source-slot matching; Section V-C also reports the unconditional accounting, which changes none of the qualitative conclusions.

VIII. Conclusion

SC-BSS of co-frequency signals is trained and ranked by waveform metrics, but the deliverable is bits. On a controlled variable-count benchmark — short bursts, comparable powers, known modulations, a fixed timing grid — we showed that a working learned separator's outputs, even under an oracle receiver, track the exact per-constellation separate-detection floor at $K \geq 2$, and we derived the receiver structure that closes the gap: a joint-MAP union bound and exact separate-detection floor sandwiching the simulated operating points, a finite-symmetry identifiability result making synchronisation and joint detection inseparable in the co-frequency regime, and an exact two-tone CRB inflation with its communication-waveform Fisher counterpart locating the disambiguating information in the modulation-bearing waveform (reachable by symbol-aware processing). The blind carrier synchroniser matches an oracle receiver within 0.6 SER points at $\text{SNR} \geq -5$ dB for BPSK/QPSK bursts (2–4 points for 8PSK), removing the oracle assumption from the evaluation pipeline at $K=1$ — and the end-to-end experiments show why that is not enough: at $K \geq 2$ independent per-slot sync is captured by the interferer's spectral line (SNR-flat +7–11 SER points, no better than no separation — reproduced with a $26\times$ larger published separator), while a training-free, ISI-aware ECM joint synchroniser-detector on the raw mixture ties the strongest waveform route at the separator quality we can reach (separate, oracle-synced, PSP/Viterbi sequence detection; 0.531 vs. 0.534) with no oracle information during inference, beats the symbol-wise oracle route at high SNR, and leaves blind mixture demodulation 11 points behind (0.643); the CNSE-built route reaches 0.444 only through its oracle synchroniser (blind: 0.536) and trails the oracle-frequency joint reference (0.411). On this benchmark and at the separator quality we can currently reach, separation serves slot generation and localisation, while detection belongs on the mixture, jointly with synchronisation — a

boundary, not a law: the true-stream control (SER 0.042) shows the joint loop would win decisively again once a separator preserves the linear mixture model faithfully enough. From the other side, blind modulation-pair classification fails at the classifier while the detector underneath it is intact, and the tested $K=3$ acquisition strategy fails by phantom-pair lock — two capabilities the tested selector and acquisition strategies do not provide. The operating envelope is $K \in \{1, 2\}$ with known modulations, and every quantitative claim is made under the considered short-burst, comparable-power, known-modulation, fixed-timing co-frequency regime. The remaining gaps are each quantified and point to concrete next steps (richer channel structure in the joint loop, and separators trained for mixture-model fidelity rather than waveform metrics).

Appendix A

Proofs of Theorems 1–2 and Propositions 1–3

Proof of Theorem 1. Conditioned on the transmitted pair $c = (c_1, c_2)$, the pairwise error event $\{c \rightarrow c'\}$ is a binary Gaussian test at distance $D = |\Delta c_1 + a_2 e^{j\varphi} \Delta c_2|$ in $\mathcal{CN}(0, \sigma^2)$ noise, hence has probability $Q(D/\sqrt{2\sigma^2})$. The event $\hat{c}_1 \neq c_1$ is the union of pairwise events with $c'_1 \neq c_1$; the union bound and averaging over the $M_1 M_2$ equiprobable hypotheses give (4). $\square$

Proof of Theorem 2. Condition on (c_2, φ) : the per-axis received value is $\ell_i + u + \nu$, $\nu \sim \mathcal{N}(0, \sigma_r^2)$, and level i is decided wrongly when it crosses the nearest boundary — above b_i with probability $Q((b_i - \ell_i - u)/\sigma_r)$ ($i < L$), below b_{i-1} with probability $Q((\ell_i + u - b_{i-1})/\sigma_r)$ ($i > 1$). Averaging the two events over the L equiprobable levels gives (6); the axes are conditionally independent, so $P_{e,\text{sep}} = \mathbb{E}[1 - (1 - e(u_r))(1 - e(u_Q))]$. For the QPSK floor at $\text{SIR} = 0$ dB: $\ell = \pm 1/\sqrt{2}$, boundary 0, $|a_2| = 1$, and with $\theta = \varphi + \arg c_2$ the offsets are $u_r = \cos \theta$, $u_Q = \sin \theta$ (signs immaterial). As $\sigma^2 \rightarrow 0$, $e(u) \rightarrow \frac{1}{2} \mathbf{1}\{|u| > 1/\sqrt{2}\}$; the events $\{|\cos \theta| > 1/\sqrt{2}\}$ and $\{|\sin \theta| > 1/\sqrt{2}\}$ are disjoint and exhaustive up to the measure-zero set $\theta \equiv \pi/4 \pmod{\pi/2}$, so almost surely exactly one axis is at risk and errs with probability $1/2: (1 - e(u_r))(1 - e(u_Q)) \rightarrow 1/2$ a.e., and bounded convergence over $\theta \sim \mathcal{U}[0, 2\pi)$ gives $P_{e,\text{sep}} \rightarrow 1/2$. On the exceptional zero-contact set both offsets equal $1/\sqrt{2}$ exactly, each axis errs with probability $Q(0) = 1/2$ per level ($e = 1/4$), and the floor drops to $1 - (3/4)^2 = 7/16$. $\square$

Proof of Proposition 1. Substitution gives $y' = e^{j(\phi_1 - \theta_1)} h_1 e^{j\theta_1} x_1 + \dots = y$ identically. $\square$

Proof of Proposition 2. Write $\mu_m^{(k)} = \mathbb{E}[c_k^m]$, $\kappa_k = \mathbb{E}|c_k|^4$, and $s = a_1 c_1 + a_2 c_2$ for the noiseless limit. For every constellation considered, rotation by $e^{j2\pi/M_k}$ is an exact set symmetry, so $\mu_m^{(k)} = 0$ unless $M_k \mid m$, while $\mu_{M_k}^{(k)} \neq 0$; also $\mathbb{E}[c_k] = 0$ and $\mathbb{E}|c_k|^2 = 1$.

Step 1 (the group acts, and no other rotation does). If $\tilde{a}_k = a_k e^{j2\pi q_k/M_k}$ then $\tilde{a}_k C_k = a_k C_k$ as sets, so the laws coincide; equal constellations add the swap. Conversely $e^{j\theta} C_k = C_k$ forces $M_k \theta \in 2\pi\mathbb{Z}$: an M -PSK set is invariant only under its own rotations, and a square-QAM grid only under multiples of $\pi/2$ (a set symmetry must permute the four extremal points).

Step 2 (amplitudes: a finite candidate set). Independence and $\mathbb{E}[c_k] = 0$ give $p := \mathbb{E}|s|^2 = |a_1|^2 + |a_2|^2$. For constellations with $\mu_2^{(k)} = 0$ (all considered except BPSK), $q := \mathbb{E}|s|^4 = \kappa_1 |a_1|^4 + \kappa_2 |a_2|^4 + 4|a_1|^2 |a_2|^2$, so $(x, y) = (|a_1|^2, |a_2|^2)$ solves $x + y = p$, $\kappa_1 x^2 + \kappa_2 y^2 + 4xy = q$. Eliminating y leaves one quadratic in x , hence at most two positive unordered solutions: the true pair $\{|a_1|^2, |a_2|^2\}$ and, when $\kappa_1 \neq \kappa_2$, possibly a spurious one (for $\kappa_1 = \kappa_2$ the two solutions are exactly the source swap). Step 2 therefore yields a finite candidate set of amplitude pairs, not a unique answer; the spurious candidate is rejected in Step 3 below. For BPSK pairs ($\mu_2 = 1$) the second circular moment $\mathbb{E}[s^2] = a_1^2 + a_2^2$ supplies the missing equation instead, again up to a finite candidate set.

Step 3 (phases). A summand of $\mathbb{E}[s^m] = \sum_j \binom{m}{j} a_1^j a_2^{m-j} \mu_j^{(1)} \mu_{m-j}^{(2)}$ survives iff $M_1 \mid j$ and $M_2 \mid (m-j)$. For unequal, nested orders $M_\ell < M_{\bar{\ell}}$, $M_\ell \mid M_{\bar{\ell}}$ (all benchmark pairs: orders 2, 4, 8), the only survivors of $\mathbb{E}[s^{M_\ell}]$ have $j \in \{0, M_\ell\}$, and $j = 0$ requires $M_\ell \mid M_{\bar{\ell}}$; hence $\mathbb{E}[s^{M_\ell}] = a_\ell^{M_\ell} \mu_{M_\ell}^{(\ell)}$.

Algorithm 1 V4: ISI-aware ECM joint synchronisation–detection

Require: mixture $r_{1:N}$; C_1, C_2 ; centred tap count L ; V1 solution $(\Delta f_1, \Delta f_2, A, c_1, c_2)$, energy $\mathcal{E}_{V1}$
1: $G \leftarrow 0_{2 \times L}$, centre taps $\leftarrow A$; $\mathcal{E} \leftarrow \mathcal{E}_{V1}$
2: repeat ▷ ISI-aware rounds, max 8
3: re-decide (c_1, c_2) by the joint grid (3) on the parallel-iterated ISI-canceled stream
4: $G \leftarrow$ closed-form LS refit of the $2L$ taps ▷ exact conditional minimiser
5: until relative energy decrease $< 10^{-4}$
6: for polish round = 1 to 4 do
7: score the 7×7 frequency grid (± 0.15 Hz, 0.05 Hz) with the $2L$ taps refit per candidate
8: move to the lowest-energy point; re-decide; re-refit; stop at relative stall $< 10^{-4}$
9: end for
10: return the lower-energy of the V4 and V1 solutions ▷ fallback; never triggered

pinning a_ℓ up to Z_{M_ℓ} . Then $\mathbb{E}[s^{M_\ell}] = a_\ell^{M_\ell} \mu_{M_\ell}^{(\ell)} + a_\ell^{M_\ell} \mu_{M_\ell}^{(\bar{\ell})}$, whose first term is invariant under the residual Z_{M_ℓ} freedom (since $M_\ell \mid M_{\bar{\ell}}$), pinning $a_{\bar{\ell}}$ up to $Z_{M_{\bar{\ell}}}$. For equal orders $M_1 = M_2 = M$, set $u = a_1^M \mu_M^{(1)}$, $v = a_2^M \mu_M^{(2)}$: $\mathbb{E}[s^M] = u + v$ and $\mathbb{E}[s^{2M}] = \alpha u^2 + \binom{2M}{M} uv + \beta v^2$ with $\alpha = \mu_{2M}^{(1)} / \mu_M^{(1)^2} \neq 0$, β likewise; eliminating u leaves one quadratic in v , so at most two ordered solutions. For $C_1 = C_2$ the system is swap-symmetric and the two solutions are exactly the S_2 pair; otherwise the spurious phase root, like the spurious amplitude candidate of Step 2, must additionally match a further known mixture moment (e.g. $\mathbb{E}[s]^6$, whose constellation factors are known and generically nonzero) — an extra algebraic equation in (a_1, a_2) that holds only on a measure-zero set and is folded into the exceptional set of Step 4. Extracting M -th roots contributes exactly the $Z_{M_1} \times Z_{M_2}$ freedom.

Step 4 (exceptional set, finiteness, jointness). If the sum map is not injective the likelihood is constant along the collision direction and the inversion fails: these are the constellation-collision rotations, exactly the zero contacts $d_{\min}(\varphi) = 0$ of Theorem 1 — a finite set, since the constellations are finite. The remaining exclusions (quadratic degeneracy, modulus ties) are algebraic equations in (a_1, a_2) , hence measure-zero; the complement is open and dense, which is the precise content of “generic”. Identifiability holds only from the joint law: per slot, a rotated constellation is itself a valid single-source hypothesis, so independent per-slot recovery cannot inherit this statement (Sections V-C–V-D). The proposition concerns the population mixture law; for finite bursts the empirical-moment inversion is consistent as $N \rightarrow \infty$, not exact. The population-law inversion is certified on 200 random generic draws for each of the ten benchmark constellation pairs by theory_identifiability.py (every recovered solution group-equivalent; collision rotations exhibit the predicted degeneracy). $\square$

Derivation for Proposition 3. Two parts. (i) The offset law. Each source’s residual offset is $\delta_k = u_k + \epsilon_k$, $u_k \sim \mathcal{U}(0, 5)$, $\epsilon_k \sim \mathcal{U}(-5, 5)$; convolving the two uniform densities gives a trapezoid on $[-5, 10]$, and the two-source separation $\Delta f = \delta_1 - \delta_2$ has the autocorrelation density of that trapezoid — piecewise quadratic on $[-15, 15]$ with knots at the multiples of 5 (closed form derived by symbolic convolution; it matches the sampler to Kolmogorov–Smirnov distance $< 2 \times 10^{-3}$ and is plotted in Fig. 3). The statistics quoted in the text follow by quadrature: $\mathbb{E}|\Delta f| = 89/24 \approx 3.708$ Hz, median 3.225 Hz, 90th/95th percentiles $7.56/8.77$ Hz, and $P(|\Delta f| < 1/T_{\text{burst}}) = 0.587$. (ii) The inflation factors. For $s_n = \sum_k A_k e^{j(2\pi f_k n / f_s + \phi_k)}$ the Fisher matrix $J_{ij} = (2/\sigma^2) \text{Re} \sum_n (\partial s_n / \partial \theta_i)^* (\partial s_n / \partial \theta_j)$ is built in closed form over $(f_1, f_2, \phi_1, \phi_2)$, resp. augmented by (A_1, A_2) , and inverted numerically; the per-tone CRB is $[J^{-1}]_{11}$ and the inflation its ratio to the single-tone value (which reproduces the Rife–Boorstyn closed form to 10^{-6} , asserted in code). The reported medians/percentiles average the exact inflation over the density of part (i) by interpolation on a 600-point offset grid. $\square$

Appendix B

The V1 procedure and the V4 algorithm

Input: symbol-rate mixture $r_{1:N}$; known constellations C_1, C_2 . Objective: $\mathcal{E}(A, \Delta f, c) = \sum_n |r_n - a_1 e^{j2\pi \Delta f_1 n T_s} c_{1,n} - a_2 e^{j2\pi \Delta f_2 n T_s} c_{2,n}|^2$. Procedure. A. Coarse search — 66 grid points (1.5 Hz over $[-5, 10]^2$ Hz, $\Delta f_1 \leq \Delta f_2$); at each, converged classification-EM on the stride-4 stream (16 gain-phase initialisations covering one symmetry sector $[0, 2\pi/M_1] \times [0, 2\pi/M_2]$); alternate joint grid decisions (3) and the LS refit $A \leftarrow RD^H(DD^H)^{-1}$ until the relative energy decrease $< 10^{-5}$, 30 rounds max). B. Fine search — 0.2 Hz grid over ± 1.2 Hz around each of the top-4 coarse points, warm-started converged classification-EM (stride 4). C. Full rate — converged classification-EM (stride 1) at the best fine point. D. Polish (≤ 6 rounds) — joint 7×7 NLS over $(\Delta f_1, \Delta f_2)$ (± 0.15 Hz, 0.05 Hz steps) with LS gain refit under the current decisions, followed by re-decision; a frequency move is accepted only at lower $\mathcal{E}$ after re-convergence; stop at relative stall $< 10^{-4}$. Multi-start solutions are ranked by $\mathcal{E}$, the minimum-energy one returned (monotonicity: Section IV-A). V4 (Algorithm 1) replaces the gain refit by a closed-form $2L$ -tap ISI refit (still an exact conditional minimiser) and the decision step by parallel-iterated ISI cancellation (only approximate, the taps coupling symbols across time) — a monotone coordinate-refinement in the generalised-EM sense, enforced by an explicit fallback to the V1 energy (never triggered in our runs).

Data availability and use of AI

All code — the generator, the baseline separator and its training script, the receiver modules, and every evaluation script behind Tables I–III — is available at https://github.com/BinNong/single_channel_source_separation/tree/main/paper6_sync_jd; the benchmark is fully synthetic and reproducible from the stated seeds. The authors used Claude Code (Anthropic) for drafting assistance and take full responsibility for the content.

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